

Homework 1.

100

- The file name of your homework (in PDF) should be in the format: “學號-作業編號.pdf”. For example: **00757999-hw1.pdf**
- Please submit your homework to Tronclass **before 23:59, October 20, 2024**

1. (10%) Order the following function by growth rate in increasing order:

$$n^2, \quad n \log(\log n), \quad n \log(n^2), \quad n^n$$

$$\text{Ans: } n \log(\log n) < n \log(n^2) < n^2 < n^n$$

2. (10%) Ackerman's function $A(m, n)$ is defined as:

$$A(m, n) = \begin{cases} n + 1, & \text{if } m = 0 \\ A(m - 1, 1), & \text{if } n = 0 \\ A(m - 1, A(m, n - 1)), & \text{otherwise} \end{cases}$$

What is the value of $A(3, 4)$?

Ans: 125

```
int Ackermann(int m, int n) {  
    if (m == 0)  
        return n + 1;  
    else if (m > 0 && n == 0)  
        return Ackermann(m - 1, 1);  
    else  
        return Ackermann(m - 1, Ackermann(m, n - 1));  
}
```

3. (12%) Show that the following statements are correct:

(a) $5n^2 - 6n = \Theta(n^2)$

(b) $n! = O(n^n)$

Ans:

(a) $5n^2 - 6n = O(n^2)$ /* $5n^2 - 6n \leq 5n^2$ for $n \geq 1$ */

$5n^2 - 6n = \Omega(n^2)$ /* $5n^2 - 6n \geq 4n^2$ for $n \geq 2$ */

$5n^2 - 6n$ 滿足 $O(n^2)$ 和 $\Omega(n^2)$ ，因此 $5n^2 - 6n = \Theta(n^2)$

(b) $n! = 1 \times 2 \times 3 \times \dots \times n$ ，而 $n^n = n \times n \times n \times \dots \times n$ 一共有 n 項。可知 $n! \leq n^n$ ，因此 $n! = O(n^n)$ 。

4. (12%) Show that the following statements are incorrect:

(a) $10n^2 + 9 = O(n)$

(b) $3^n = O(2^n)$

Ans:

(a) Big O 符號定義：

(a) if $3^n = O(2^n)$, there exist a constant $c > 0$ and $n_0 > 0$ such that $3^n \leq c \cdot 2^n$ for $n \geq n_0$. But $3^n \geq (c+1)2^n$ for $n \geq \log_{3/2}(c+1)$. So, if $n \geq \max(n_0, \log_{3/2}(c+1))$, $(c+1)2^n \leq 3^n \leq c \cdot 2^n$ which means $1 \leq 0$ (✗)

$f(n)=O(g(n))$ 的意思是存在常數 c 和 n_0 ，使得對於所有 $n \geq n_0$ ，有 $f(n) \leq cg(n)$

平方項比線性項增長得更快，因此 $10n^2 + 9$ 不能被 $O(n)$ 上界

(b) 3^n 和 2^n 是指數函數，但底數不同。因為 $3 > 2$ ，增長速度比較快。隨著 n 增加，兩者的比率會呈指數增長，增大到趨近於無窮大。因此，不可能有一個常數 c 使得 $3^n \leq c \cdot 2^n$ 對所有 $n \geq n_0$ 成立。

因此 3^n 不能被 $O(2^n)$ 上界。

5. (16%) (from NTHU 106) Which ones of the following conjectures are correct? Please justify your answers:

(a) $3n^2 - 100n + 6 = O(n)$

(b) $\log(n!) = \Theta(n \log n)$

(c) $n! = O(n^n)$

(d) $(n+a)^b = \Theta(n^b)$ where a and b are real constants

(e) $(n+a)^b = \Omega(n^{b+1})$ where a and b are real constants

Ans: (b), (c), (d)

(a) 錯誤， $3n^2 - 100n + 6 \neq O(n)$ ， $c \cdot n < 3n^2$ 當 $n > c$

(b) 正確，

$$\log(n!) = O(n \log n) : \log(1) + \log(2) + \dots + \log(n) \leq \log(n) + \log(n) + \dots + \log(n) = n \cdot \log(n)$$

$$\log(n!) = \Omega(n \log n) : \log(1) + \dots + \log(n/2) + \dots + \log(n) \geq \log(n/2) + \dots + \log(n) = \log(n/2) + \log(n/2+1) + \dots + \log(n-1) + \log(n) \geq \log(n/2) + \dots + \log(n/2) = n/2 \cdot \log(n/2)$$

$$\text{可得 } \log(n!) = \Theta(n \log n)$$

(c) 正確， $n! = n \cdot (n-1) \cdot (n-2) \cdot (n-3) \cdot \dots \cdot 3 \cdot 2 \cdot 1 \leq n^n = n \cdot n \cdot n \cdot \dots \cdot n \cdot n \cdot n$

(d) 正確，

$$n \geq |a|, n+a \leq n+|a| \leq 2n$$

$$n \geq 2|a|, n+a \geq n-|a| \geq \frac{1}{2}n$$

因此，當 $n \geq 2|a|$ ， $0 \leq \frac{1}{2}n \leq n+a \leq 2n$ 。

由於 b is a positive constant:

$$0 \leq (\frac{1}{2}n)^b \leq (n+a)^b \leq (2n)^b$$

$$0 \leq (\frac{1}{2})^b n^b \leq (n+a)^b \leq (2^b) n^b$$

因此， $c_1 = (\frac{1}{2})^b$ 、 $c_2 = 2^b$ 、 $n_0 = 2|a|$ 滿足定義

(e) 錯誤，

$$(n+a)^b \geq c \cdot n^{b+1} \text{ 成立時，} (n+a)^b = \Omega(n^{b+1}) \text{ 才正確}$$

$$\text{當 } n \text{ 很大時，} (n+a)^b \approx n^b + ba \cdot n^{b-1}$$

根據近似值，當 n 趨近無窮大時， $(n+a)^b$ 的增長速率為 n^b

$(n+a)^b$ 的增長速率比 n^{b+1} 慢，無法找到一個常數 c 使得 $(n+a)^b \geq c \cdot n^{b+1}$ 對所有大的 n 成立。

6. (20%) Fill the step count table (yellow regions) for the following program:

Statement	s/e	Frequency	Total Steps
void mult(int a[][MAX_SIZE],int b[][MAX_SIZE], int c[][MAX_SIZE])	0	0	0
{	0	0	0
int i,j, k;	0	0	0
for (i = 0; i < MAX_SIZE; i++)	1	MAX_SIZE+1	MAX_SIZE+1
for (j = 0; j < MAX_SIZE; j++) {	1	$MAX_SIZE(MAX_SIZE+1)$	$MAX_SIZE+MAX_SIZE^2$
c [i][j] = 0;	1	MAX_SIZE^2	MAX_SIZE^2
for (k = 0; k < MAX_SIZE; k++)	1	$MAX_SIZE^2(MAX_SIZE+1)$	$MAX_SIZE^2(MAX_SIZE+1)$
c [i][j] += a [i][k] * b [k][j];	1	MAX_SIZE^3	MAX_SIZE^3
}	0	0	0
}	0	0	0
Total			$2*MAX_SIZE^3+3*MAX_SIZE^2+2*MAX_SIZE+1$

7. (10%) Given an four-dimensional array A[100][200][300][400], if the address of A[3][20][32][45] is $\alpha + X$, what is the value of X ?

Note: (1) α is the address of A[0][0][0][0]

(2) Only consider row major here.

Ans: $X=45+32 \times 400+20 \times 300 \times 400+3 \times 200 \times 300 \times 400=74412845$

8. (10%) (from NTCU 107) Consider the following function. Please derive the time complexity of function foo using recurrence relation. Please show your derivation step by step.

```

int i[n]; /* Assume that the values of all integers in i[n] have been initialized */
int foo(int a, int b, int c) {
    int d;
    if (a>b) return -1;
    d=(a+b)/2;
    if (c==i[d])
        return d;
    else if (c>i[d])
        return foo(d+1,b,c);
    else
        return foo(a, d-1,c);
}

```

Ans:

Statement	s/e	Frequency	Total steps
int foo(int a, int b, int c)	0	0	0
{	0	0	0
int d;	1	1	1
if (a > b) return -1;	1	1	1
d = (a + b) / 2;	1	1	1
if (c == i[d]) return d;	1	1	1
else if (c > i[d]) return foo(d+1, b, c);	1	$T(n/2)$	$T(n/2)$
else return foo(a, d-1, c);	1	$T(n/2)$	$T(n/2)$
}	0	0	0
total			$T(n) = O(\log n)$

每次遞迴步驟中，搜尋空間會被減半，函式的遞迴關係：

$$T(n) = T(n/2) + O(1)$$

每次遞迴呼叫時，問題規模減半，遞迴呼叫中的常數時間操作是 $O(1)$

可知 $T(n) = O(\log n)$